

- 27 What is the de Broglie wavelength of an alpha particle of kinetic energy E , given that u is the unified atomic mass unit?

A $\frac{h}{2\sqrt{uE}}$

B $\frac{2\sqrt{uE}}{h}$

C $\frac{h}{2\sqrt{2uE}}$

D $\frac{2\sqrt{2uE}}{h}$

$$E = \frac{1}{2} m v^2 = \frac{p^2}{2m} \rightarrow p = (2mE)^{1/2} = (8uE)^{1/2}$$

$$\lambda = h / p = h / (8uE)^{1/2}$$

28 The beta spectrum for ^{12}B decay is as shown below